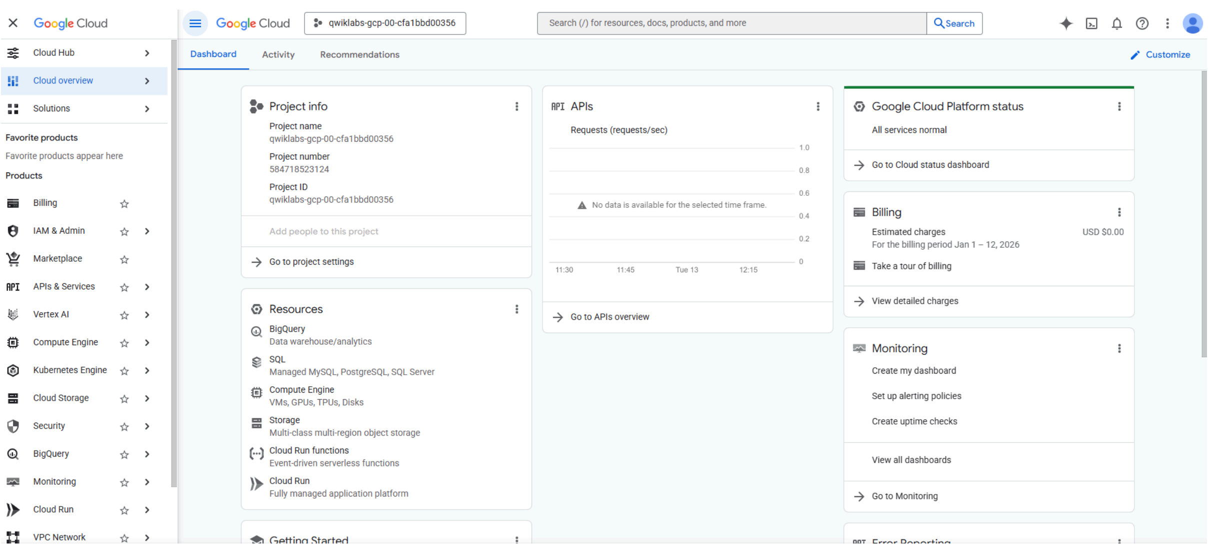
Compute Engine: Qwik Start Windows

Compute Engine lets you create and run virtual machines on Google infrastructure. Compute Engine offers scale, performance, and value that allows you to easily launch large compute clusters on Google's infrastructure.

You can run your Windows applications on Compute Engine and take advantage of many benefits available to virtual machine instances, such as reliable storage options, the speed of the Google network, and Autoscaling.

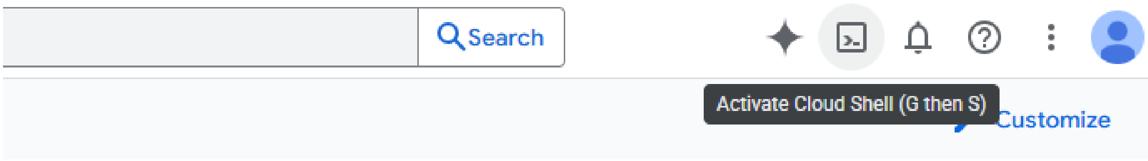
In this hands-on lab, you learn how to launch a Windows Server instance in Compute Engine and use Remote Desktop Protocol (RDP) to connect to it.

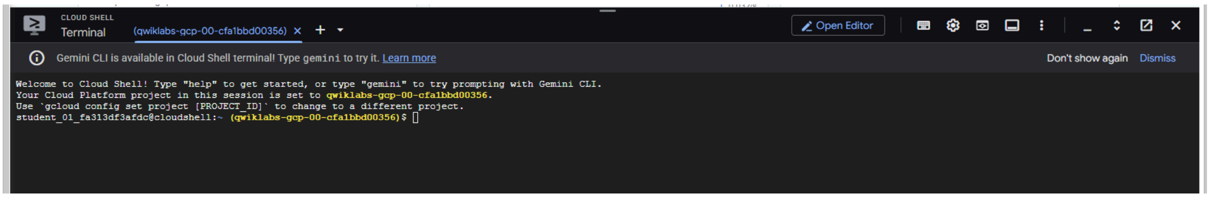


Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

1. Click **Activate Cloud Shell** at the top of the Google Cloud console.





1. Click through the following windows:
    - Continue through the Cloud Shell information window.
    - Authorize Cloud Shell to use your credentials to make Google Cloud API calls.

When you are connected, you are already authenticated, and the project is set to your **Project_ID**, `PROJECT_ID`. The output contains a line that declares the **Project_ID** for this session:

`gcloud` is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

1. (Optional) You can list the active account name with this command:

```
gcloud auth list
```

Copied!

1. Click **Authorize**.

Output:

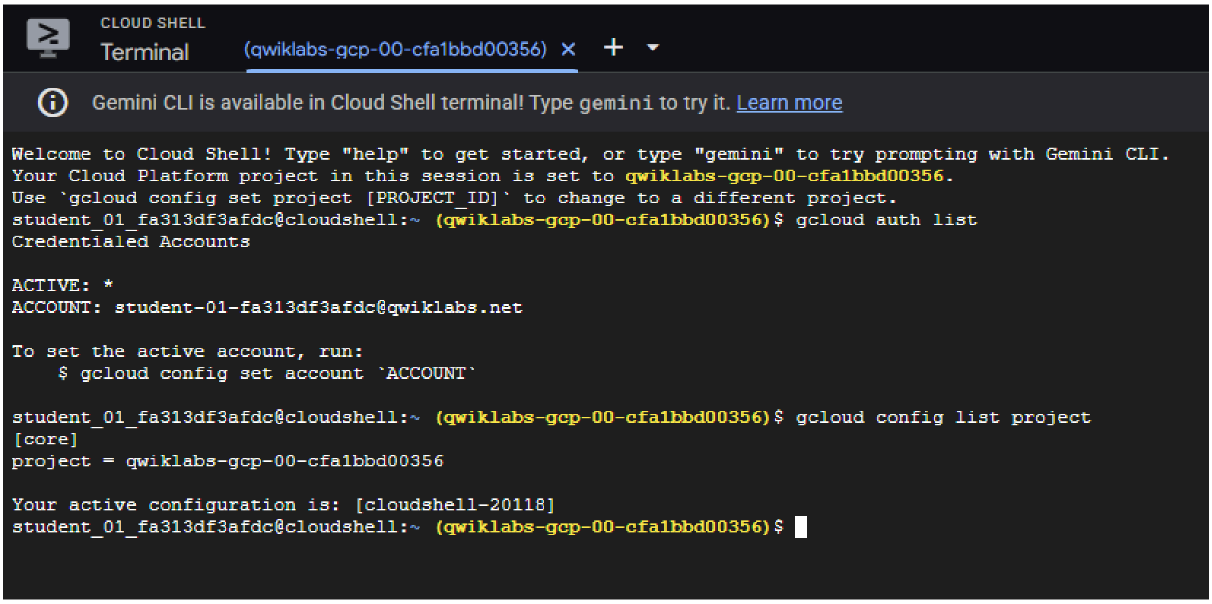
```
ACTIVE: *
ACCOUNT: "ACCOUNT"

To set the active account, run:
    $ gcloud config set account `ACCOUNT`
```

1. (Optional) You can list the project ID with this command:

```
gcloud config list project
```

Output:

```
[core]
project = "PROJECT_ID"
```



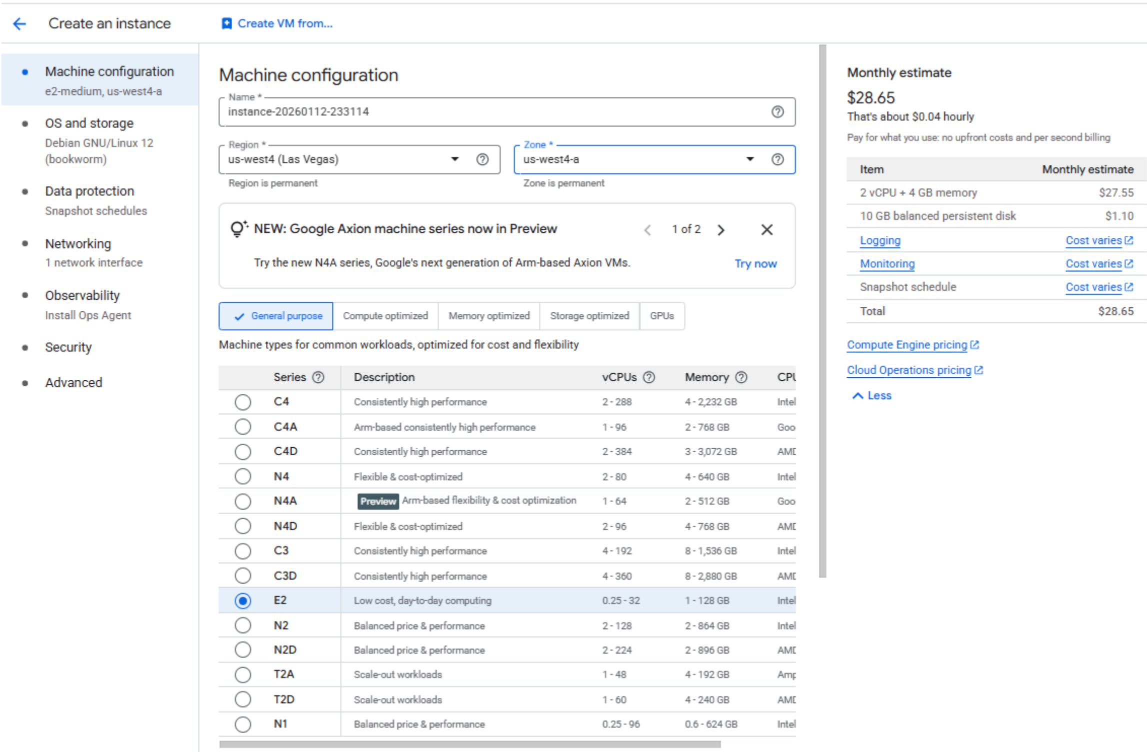
Create a virtual machine instance

1. In the Cloud Console, on the **Navigation menu** (), click **Compute Engine > VM instances**, and then click **Create Instance**.

2. In the **Machine configuration**.

   Select the following values:

Property	Value (type value or select option as specified)
Region	us-west4
Zone	us-west4-a
Series	E2



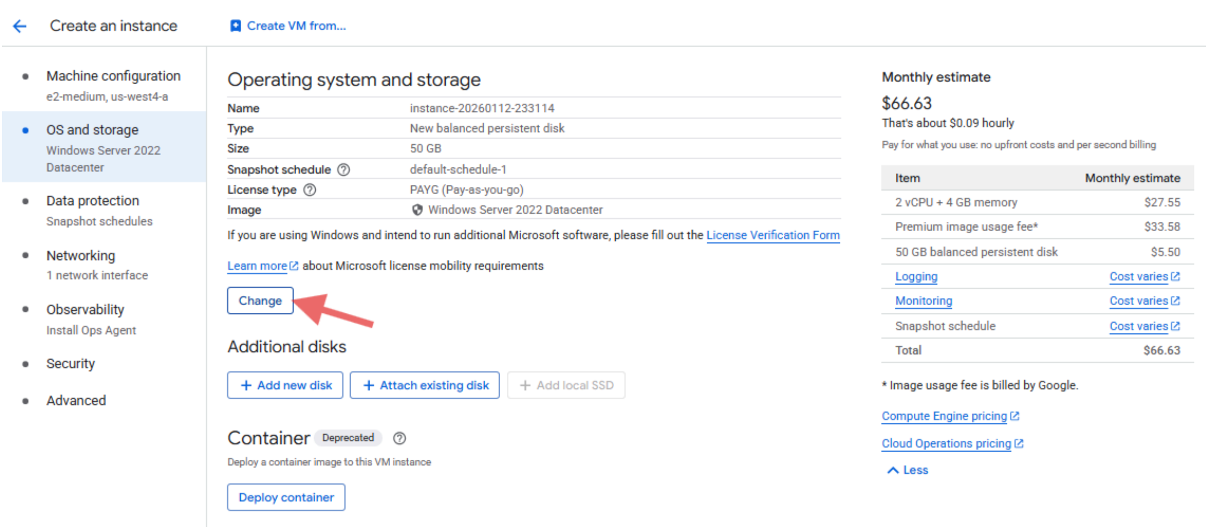
3. Click **OS and storage**.

   Click **Change** to begin configuring your boot disk and select the following values:

   - **Operating system**: `Windows Server`
   - **Version**: `Windows Server 2022 Datacenter`

   Click **Select**.

4. Click **Create** to create the instance.



Remote Desktop (RDP) into the Windows Server

Test the status of Windows Startup

After a short time, the Windows Server instance will be provisioned and listed on the VM Instances page with a green status icon.

The server instance may not be ready to accept RDP connections, as it takes a while for all OS components to initialize.

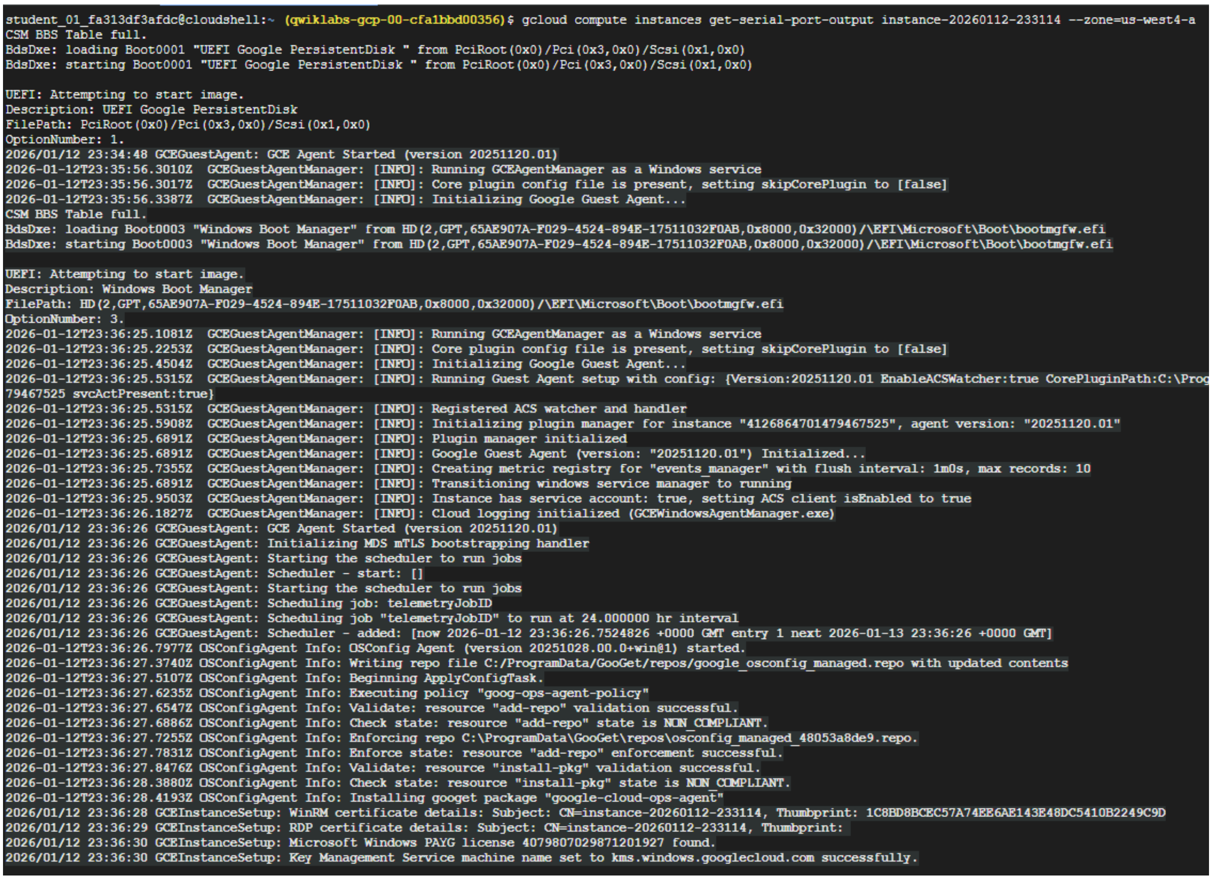
1. To see whether the server instance is ready for an RDP connection, run the following command at your Cloud Shell terminal command line and please make to replace `[instance]` with the VM Instance that you created earlier.

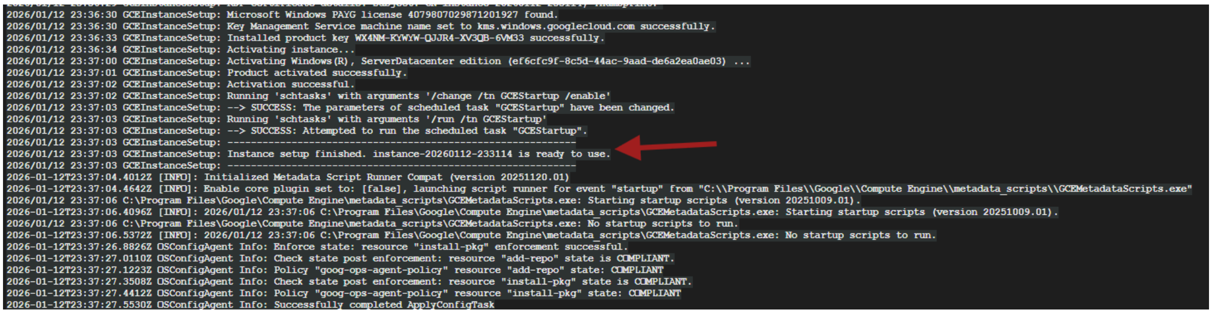
```
gcloud compute instances get-serial-port-output [instance] --zone=ZONE
```

1. If prompted, type N and press ENTER.

Repeat the command until you see the following in the command output, which tells you that the OS components have initialized and the Windows Server is ready to accept your RDP connection.

```
------------------------------------------------------------
Instance setup finished. instance is ready to use.
------------------------------------------------------------
```





RDP into the Windows Server

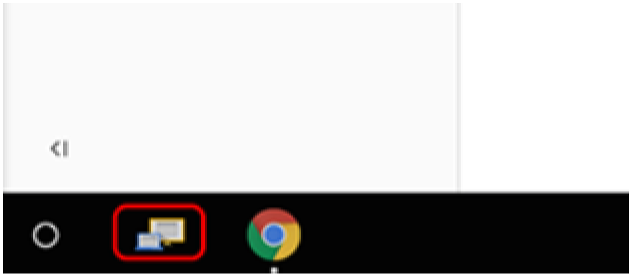
1. To set a password for logging into the RDP, run the following command in Cloud Shell. Be sure you replace `[instance]` with the VM Instance that you created and set `[username]` as **admin**.

```
gcloud compute reset-windows-password [instance] --zone ZONE --user [username]
```

1. If asked `Would you like to set or reset the password for [admin] (Y/n)?`, enter Y. Record the password for use in later steps to connect.

2. Connect to your server. There are different ways to connect to your server through RDP, depending on whether you are on Windows or not:

   - If you are using a Chromebook or other machine at a Google Cloud event there is likely an RDP app already installed on the computer. Click the icon as below, if it is present, in the lower left corner of the screen and enter the external IP of your VM.



1.

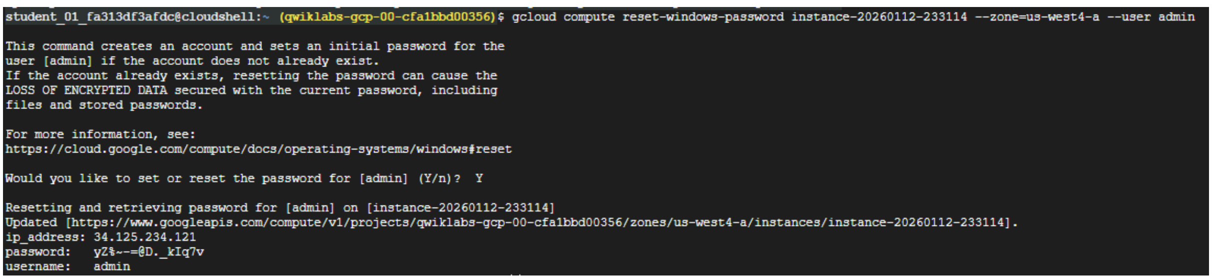
   - If you are not on Windows but using Chrome, you can connect to your server through RDP directly from the browser using the [Spark View]() extension. Click on **Add to Chrome**. Then, click **Launch app**.

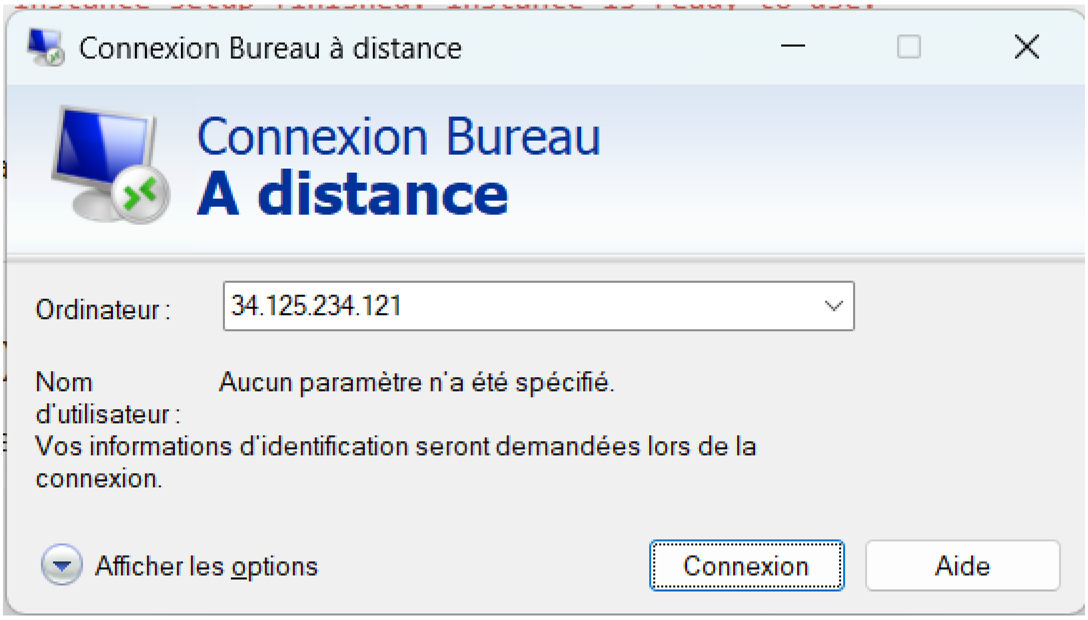
2. Once launched, the **Spark View (RDP)** window opens. Use your Windows username **admin** and password you previously recorded in Step 2.

3. Add your VM instance's External IP as your Domain. Click Connect to confirm you want to connect.

If you are on a Macintosh, there are several freely accessible RDP Client packages available to install, such as [CoRD](). After installing, connect as above to the External IP address of the Windows server. Once it has connected, it will open up a login page where you can specify Windows username **admin** and password from the output of above mentioned command to log in (ignore the Domain: field).

Once logged in, you should see the Windows desktop!



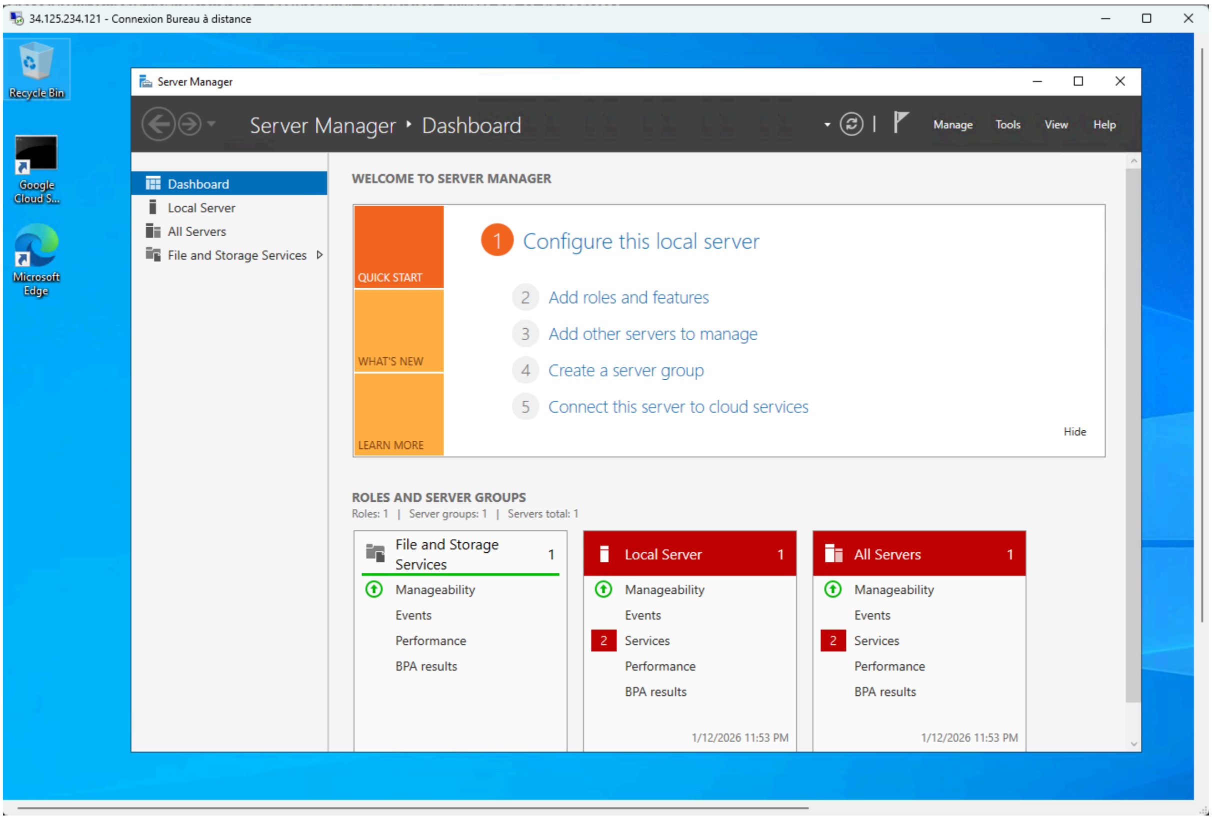


Copy and paste with the RDP client

Once you are securely logged into your instance, you may find yourself copying and pasting commands from the lab manual.

To paste, hold the **CTRL-V** keys (if you are a Mac user, using **CMND-V** will not work.) If you are in a Powershell window, be sure that you have clicked into the window or else the paste shortcut won't work.

If you are pasting into putty, **right click**.



Une VM est comme un ordinateur classique jusque que celui-ci vie dans un datacenter de chez google et on y accede via le reseau

Dans ce lab en selectionnant windows server 2022 data center comme OS cela fait que notre VM va se comporter comme un server physique qui va run programme

une fois que nous avons installer la VM est que celle-ci est ready to use comment bien verifier cela on utiliser la commande et cela va nous renvoyer plein d'information technique et une fois que nous observons le message comme quoi l'iinstance est ready cela est good

```
gcloud compute instances get-serial-port-output [instance] --zone=ZONE
```

Nous avons besoin de nous connecter a notre interface donc nous allons creer une mdp pour notre user admin en utilisant la commande , une fois que nous avons le mdp et le user nous allons utiliser le protocole (RDP Remote Desktop Protocol) qui est le bureau distant et cela va ouvrir Windows Server